Discretized Quantum Adiabatic Evolution to Solve QUBO problems

Hrishikesh Belagali

This manuscript was compiled on June 4, 2025

Abstract

This project proposed a quantum adiabatic evolution to solve Quadratic Binary Unconstrained Optimization (QUBO) problems. The problems were formulated as quantum Ising lattices with a time-dependent transverse magnetic field. The time evolution operator was discretized using the Trotter Suzuki approximation. The algorithm was implemented in Python using IBM's Qiskit quantum developmental package.

Keywords: *QUBO, Ising, Quantum Adiabatic Evolution, Quantum Computing, Optimization*

Rho LaTeX Class This document is licensed under Creative Commons CC BY 4.0.

1. Introduction

The origins of quantum computing can be traced back to quantum Turing machines (Benioff, 1980) and an ambition to resolve the difficulties associated with simulating quantum mechanics on classical computers. The standard model of quantum computing is the gate model (Deutsch & Penrose, 1989), in which computations are carried out using series of unitary quantum logic gates. Adiabatic Quantum Computation (AQC) is fundamentally different from the standard model. In AQC, computations are initiated from an initial Hamiltonian with a trivial ground state, to a final Hamiltonian whose ground state represents the solution to the desired problem (Albash & Lidar, 2018). The adiabatic theorem ensures that the evolving Hamiltonian will remain in its instantaneous ground state throughout the evolution, given that the evolution is carried out slowly.

The earliest form of AQC was quantum stochastic optimization (Apolloni et al., 1989), which was used to solve classical combinatorial optimization problems. It was later renamed quantum annealing (QA) (Apolloni et al., 1988). The initial motivation for quantum annealing was to be understood as a quantum-inspired variant of simulated annealing, which utilizes simulated quantum fluctuations and tunneling rather than thermal (Kirkpatrick et al., 1983). The two algorithms were later compared and it was verified that quantum annealing could be more powerful (Kadowaki & Nishimori, 1998). The experimental implementation of QA in a disordered quantum ferromagnet (Brooke et al., 1999) allowed for the algorithm to be considered from the perspective of quantum computing, involving a designated quantum device.

Adiabatic quantum computation generally involves stoquastic Hamiltonians (Albash & Lidar, 2018), which have non-positive off-diagonal elements in the standard computational basis. Non-stoquastic AQC can be simulated by all other universal quantum computation techniques within polynomial resource overhead (Albash & Lidar, 2018)

Quadratic Unconstrained Binary Optimization (QUBO) is a classical combinatorial optimization problem with widespread applications. Given a matrix $\mathbf{Q}$ of order n, the QUBO problem is determining

$$\mathbf{x}^* = \underset{\mathbf{x} \in \{0,1\}^n}{\arg\min}\, \mathbf{x}^T Q \mathbf{x}$$

In other words, the aim is to determine the bit vector corresponding to the minimum of the quadric form. It is an NP-hard problem, eliciting the need for metaheuristic algorithms. The aim of this project is to formulate QUBO problems as quantum Ising lattice Hamiltonians and solve them using AQC.

2. Theory

As mentioned, AQC works by adiabatically evolving from an initial Hamiltonian $\hat{H}_i$ with a trivial ground state to the problem Hamiltonian $\hat{H}_p$. The time evolution operator is discretized using the Trotter-Suzuki approximation.

2.1. Problem Formulation

The classical Ising model has been used as a formulation tool for notable NP-hard problems such as maximum graph cut optimization (Cook et al., 2019). It consists of discrete spin variables σ which store atomic spin values. The model is useful for identifying phase transitions. It is given by the following equation-

$$\hat{H} = -\sum_{i=1}^{n}\sum_{j=1}^{n} J_{ij}\sigma_i\sigma_j - \sum_{i=1}^{n} h_i\sigma_i$$

where $\mathbf{J}$ represents the interaction energy matrix and $\mathbf{h}$ represents a vector of coupling coefficients.

2.1.1. Quantum Ising Model

The quantum Ising model is fundamental in quantum mechanics and statistical physics. It represents an n-particle quantum system wherein the z-aligned spin of each particle is determined by interactions with other particles (Lucas, 2014). There is also an orthogonal external magnetic field which energetically favors one x-axis spin direction over the other. The non-commuting nature of Z and X spin observables requires a quantum approach.

$$\hat{H} = -\sum_{i=1}^{n}\sum_{j=1}^{n} J_{ij}\mathcal{S}_i^Z\mathcal{S}_j^Z - \sum_{i=1}^{n} h_i\mathcal{S}_i^X$$

where

$$\mathcal{S}_i^Z = \mathbf{I}_{2^i} \otimes \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \otimes \mathbf{I}_{2^{(n-i-1)}}$$

$$\mathcal{S}_i^X = \mathbf{I}_{2^i} \otimes \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \otimes \mathbf{I}_{2^{(n-i-1)}}$$

Note that $\mathbf{I}_n$ is the n-order identity matrix, and $\otimes$ is the tensor product. The 2^n order matrices shown above are controlled Pauli matrices operating in n-particle multipartite Hilbert space.

2.1.2. Quantum Ising Model Formulation of QUBO

QUBO problems can be formulated as classical Ising Hamiltonians through simple, linear substitutions. From there, building the quantum variant is a matter of replacing spin variables with 2^n-order controlled Pauli matrices. This substitution is permissible because the eigenvalues of the matrices are equivalent to the states of the discrete spin variables.

Let the QUBO problem be $f(\mathbf{x}) = \mathbf{x}^T Q\mathbf{x}$. It can be rewritten in the following form-

$$f(\mathbf{x}) = \sum_{i=1}^{n}\sum_{j=1}^{n} Q_{ij}x_i x_j$$

Let $\boldsymbol{\sigma} = 2\mathbf{x} - 1$, where $\boldsymbol{\sigma}$ is a vector of spin variables.

$$f(\boldsymbol{\sigma}) = \sum_{i=1}^{n}\sum_{j=1}^{n} Q_{ij}\left(\frac{\sigma_i+1}{2}\right)\left(\frac{\sigma_j+1}{2}\right) \tag{1}$$

$$= \sum_{i=1}^{n}\sum_{j=1}^{n}\frac{Q_{ij}}{4}\sigma_i\sigma_j + \sum_{i=1}^{n}\sum_{j=1}^{n}\frac{Q_{ij}(\sigma_i+\sigma_j)}{4} + \sum_{i=1}^{n}\sum_{j=1}^{n}\frac{Q_{ij}}{4} \tag{2}$$

$$= \sum_{i=1}^{n}\sum_{j=1}^{n}\frac{Q_{ij}}{4}\sigma_i\sigma_j + \sum_{i=1}^{n}\left(\sum_{j=1}^{n}\frac{Q_{ij}+Q_{ji}}{4}\right)\sigma_i + \sum_{i=1}^{n}\sum_{j=1}^{n}\frac{Q_{ij}}{4} \tag{3}$$

$$= -\sum_{i=1}^{n}\sum_{j=1}^{n} J_{ij}\sigma_i\sigma_j - \sum_{i=1}^{n} h_i\sigma_i + C \tag{4}$$

where

$$J_{ij} = \frac{-Q_{ij}}{4}$$

$$h_i = -\left(\sum_{j=1}^{n}\frac{Q_{ij}+Q_{ji}}{4}\right)$$

$$C = \sum_{i=1}\sum_{j=1}\frac{Q_{ij}}{4}$$

The constant can be neglected as it does not affect the optimization process. By replacing the spin variables with controlled Pauli matrices, we get a quantum Ising Hamiltonian -

$$\hat{H}_p = -\sum_{i=1}^{n}\sum_{j=1}^{n} J_{ij}\mathcal{S}_i^Z\mathcal{S}_j^Z - \sum_{i=1}^{n} h_i\mathcal{S}_i^X$$

This serves as the problem Hamiltonian for use in AQC.

2.1.3. The Initial Hamiltonian

The initial Hamiltonian must be constructed to be noncommuting with the final Hamiltonian -

$$[\hat{H}_i, \hat{H}_p] \neq 0$$

The ground state of the adiabatic Hamiltonian must remain nondegenerate throughout the evolution. In other words, anti-crossing must be ensured for the ground state and a finitely-large spectral gap must exist between the ground state and first excited state at all times. This will be elaborated on more in Section 2.2. Through empirical testing, the following initial Hamiltonian was found to satisfy the given conditions, assuming sufficient gap between the minima of the QUBO problem.

$$\hat{H}_i = \sum_{i=1}^{n}\mathcal{S}_i^Z$$

The ground state is prepared as

$$|\Psi\rangle_0 = \bigotimes_{i=1}^{n}\mathcal{H}_i\,|0\rangle_i$$

where $\mathcal{H}_i$ is the Hadamard matrix acting on the ith particle (qubit).

$$\mathcal{H} = \frac{1}{\sqrt{2}}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix}$$

The provided superposition state will provide ample exploration of the solution space.

2.2. Time Evolution

2.2.1. The Adiabatic Hamiltonian

The adiabatic Hamiltonian is a smooth interpolation between the initial and final Hamiltonian (Lucas, 2014).

$$\hat{H}(t) = (1 - s(t))\cdot\hat{H}_i + s(t)\cdot\hat{H}_f$$
$$s : [0, T] \to [0, 1]$$

2.2.2. The Adiabatic Evolution & Trotter-Suzuki Approximation

The adiabatic condition is approximated by the following relation -

$$\frac{\hbar|s'(t)|}{\Delta E_{min}^2} \ll 1$$

For the purposes of our implementation, the adiabatic frequency is taken to be few orders of magnitude less than the intrinsic frequency of the quantum system.
Traditionally, the Dyson operator must be used for the time evolution of a time-dependent Hamiltonian $\hat{H} = \hat{H}_0 + \hat{V}_S(t)$.

$$|\Psi(t)\rangle = \sum_{n=0}^{\infty}\frac{(-i\hbar^{-1})^n}{n!}\underbrace{\int \mathrm{d}t_1\cdots\mathrm{d}t_n}_{t_f\geq t_1\cdots\geq t_n\geq t_i}\mathcal{T}\left\{\prod_{k=1}^{n} e^{\frac{iH_i t_k}{\hbar}}V(t_k)e^{-\frac{iH_i t_k}{\hbar}}\right\}|\Psi(t_0)\rangle$$

where $\mathcal{T}$ is the time-ordering operator. However, it is computationally expensive to implement. Instead, we discretize the evolution operator using the Trotter-Suzuki approximation.

$$U(0,T) = e^{-iHT} = e^{-iH\frac{T}{n}\cdot n} \approx \left(e^{-iH\delta t}\right)^n + \mathcal{O}(\delta t)$$

where $n\delta t = T$. The error arises due to the noncommuting nature of the discrete Hamiltonians.

Note that as $n \to \infty$, $\mathcal{O}(\delta t) \to 0$. Hence, states can be evolved discretely as

$$|\Psi(t+\delta t)\rangle = U(t, t+\delta t)\,|\Psi(t)\rangle$$

The motivation for this approximation is that as $n \to \infty$, the Hamiltonian's change with respect to time becomes less significant, so the error from using the time-independent evolution operator decreases.

2.2.3. The Anticrossing Problem

The adiabtic theorem is applicable only if the instantaneous eigenstates of the Hamiltonian remain nondegenerate throughout the evolution. In other words, the energy values of two or more eigenstates should not be equal throughout the transition. Ample proximity between eigenstates can be ensured through slower evolution rates, however the adiabatic Hamiltonian must be engineered in such a way that the eigenstates do not cross over.

3. Methods

The algorithm was implemented using Python 3.11.9, Qiskit 1.3.1 and NumPy 1.26.4. It was tested on small QUBO problems using the Aer noisy quantum simulator. The complete implementation of the algorithm can be found here.

3.1. Challenges and Future Work

As mentioned in Section 2.2.3, the algorithm requires nondegenerate eigenstates throughout the course of evolution. However, for certain near-equal values in the QUBO matrix, the adiabatic Hamiltonian used in this implementation can occassionally cause a loss of nondegeneracy, resulting in erroneous optimization. Although this can be mitigated through upscaling, further work will engineer sufficient gaps between the Hamiltonian eigenstates.

As with many quantum algorithms, large-scale classical simulations are difficult due to the exponential scaling of n-particle Hilbert space. Therefore, further work will also look into implementations that are feasible for larger QUBO matrices.

References

Albash, T., & Lidar, D. A. (2018). Adiabatic quantum computation. *Reviews of Modern Physics*, *90*(1). https://doi.org/10.1103/revmodphys.90.015002

Apolloni, B., Carvalho, C., & de Falco, D. (1989). Quantum stochastic optimization. *Stochastic Processes and their Applications*, *33*(2), 233–244. https://doi.org/https://doi.org/10.1016/0304-4149(89)90040-9

Apolloni, B., Cesa-Bianchi, N., & de Falco, D. (1988). Proceedings of the ascona/locarno conference, p. https://cesa-bianchi.di.unimi.it/Pubblicazioni/quantumAnnealing.pdf

Benioff, P. (1980). The computer as a physical system: A microscopic quantum mechanical hamiltonian model of computers as represented by turing machines. *Journal of Statistical Physics*, *22*(5), 563–591. https://doi.org/10.1007/BF01011339

Brooke, J., Bitko, D., F., T., null, & Aeppli, G. (1999). Quantum annealing of a disordered magnet. *Science*, *284*(5415), 779–781. https://doi.org/10.1126/science.284.5415.779

Cook, C., Zhao, H., Sato, T., Hiromoto, M., & Tan, S. X.-D. (2019). Gpu-based ising computing for solving max-cut combinatorial optimization problems. *Integration*, *69*, 335–344. https://doi.org/https://doi.org/10.1016/j.vlsi.2019.07.003

Deutsch, D. E., & Penrose, R. (1989). Quantum computational networks. *Proceedings of the Royal Society of London. A. Mathematical and Physical Sciences*, *425*(1868), 73–90. https://doi.org/10.1098/rspa.1989.0099

Kadowaki, T., & Nishimori, H. (1998). Quantum annealing in the transverse ising model. *Phys. Rev. E*, *58*, 5355–5363. https://doi.org/10.1103/PhysRevE.58.5355

Kirkpatrick, S., Gelatt, C. D., & Vecchi, M. P. (1983). Optimization by simulated annealing. *Science*, *220*(4598), 671–680. https://doi.org/10.1126/science.220.4598.671

Lucas, A. (2014). Ising formulations of many np problems. *Frontiers in Physics*, *Volume 2 - 2014*. https://doi.org/10.3389/fphy.2014.00005